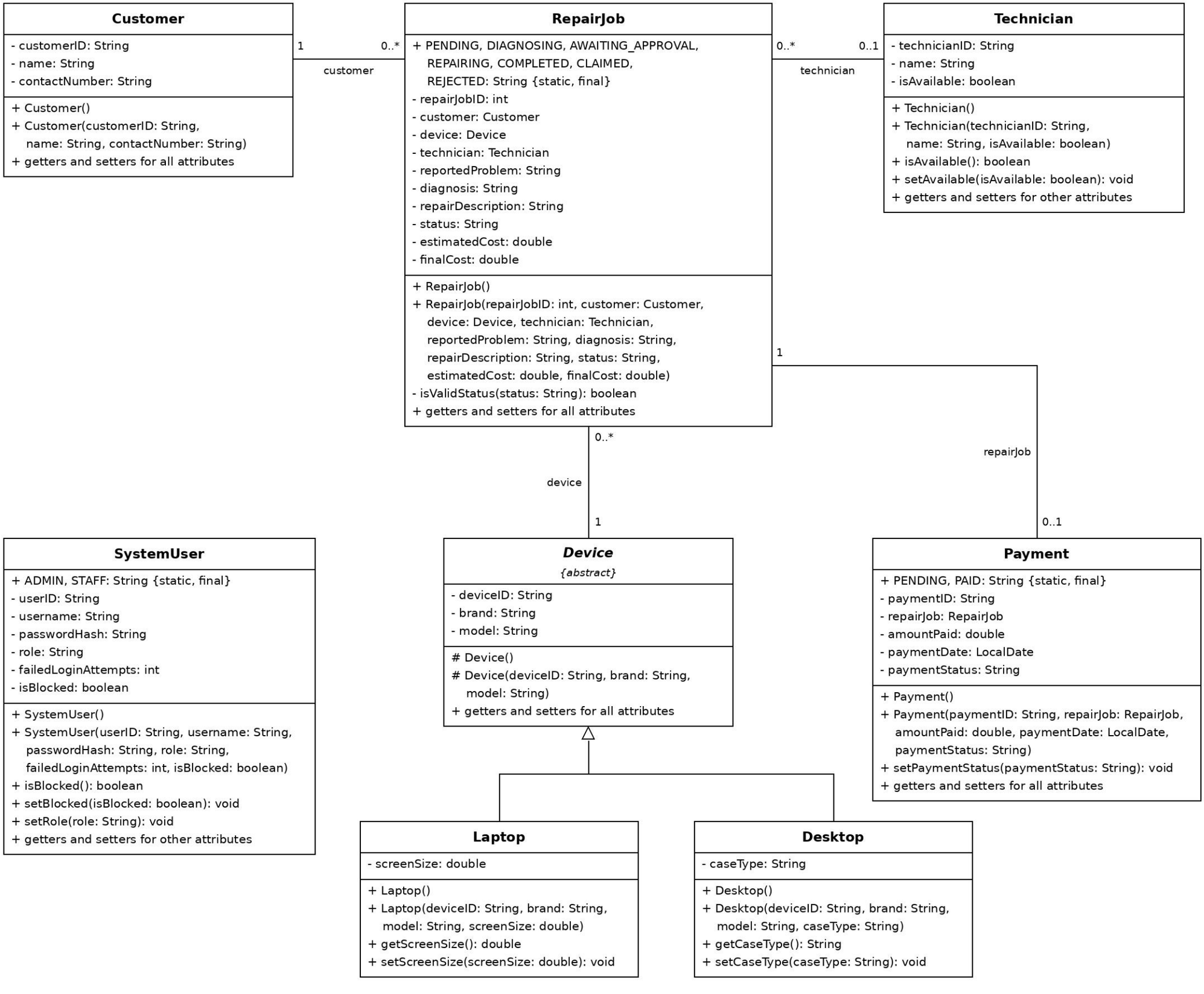
Computer Repair Shop Service System - UML Class Diagram (Domain Classes)



Computer Repair Shop Service System - UML Class Diagram (Application Classes)

Main

+ main(args: String[]): void {static}

«create»

LoginFrame

extends JFrame

- dataManager: DataManager
- usernameField: JTextField
- passwordField: JPasswordField
- signInButton: JButton
- statusLabel: JLabel

+ LoginFrame(dataManager: DataManager)
- createContent(): JPanel
- fieldLabel(text: String): JLabel
- styleField(field: JTextField): void
- signIn(): void

«create»

MainFrame

extends JFrame

- dataManager: DataManager
- authenticatedUser: SystemUser
- contentPanel: JPanel
- navigationButtons: Map<String, JButton>
- currentPage: String

+ MainFrame(dataManager: DataManager, authenticatedUser: SystemUser)
- createSidebar(): JPanel
- createNavigationButton(text: String): JButton
- navigateTo(page: String): void
- showDashboard(): void
- showCustomers(): void
- showDevices(): void
- showRepairJobs(): void
- showTechnicians(): void
- showPayments(): void
- showUsers(): void
- showNotImplemented(page: String): void
- updateNavigationSelection(): void
- logout(): void

«create»

Main «create» LoginFrame

LoginFrame 1 — dataManager

Main «create» DataManager

DataManager

- customers: ArrayList<Customer>
- devices: ArrayList<Device>
- technicians: ArrayList<Technician>
- repairJobs: ArrayList<RepairJob>
- payments: ArrayList<Payment>
- systemUsers: ArrayList<SystemUser>
- waitingQueue: Queue<RepairJob>
- currentAuthenticatedUserID: String
- customerCounter, deviceCounter, technicianCounter, repairJobCounter, paymentCounter, userCounter: int
- secureRandom: SecureRandom
- HASH_ITERATIONS, SALT_LENGTH, HASH_LENGTH: int {static, final}

+ DataManager()

Authentication and session

+ authenticate(username, password): SystemUser
+ logout(): void
+ getCurrentAuthenticatedUserID(): String
+ initializeDefaultAccounts(): boolean

Users

+ createStaffAccount(actorUserID, username, password): SystemUser
+ updateUsername(actorUserID, userID, newUsername): boolean
+ changePassword(...): boolean
+ unblockUser(actorUserID, userID): boolean
+ deleteSystemUser(actorUserID, userID): boolean
+ getSystemUsers(): List<SystemUser>
+ getSystemUserByID(id) / getSystemUserByUsername(username)

Customers

+ addCustomer(actorUserID, name, contactNumber): Customer
+ updateCustomer(actorUserID, customerID, name, contactNumber): boolean
+ deleteCustomer(actorUserID, customerID): boolean
+ getCustomers(): List<Customer>
+ getCustomerByID(id): Customer

Devices

+ addDesktop(actorUserID, brand, model, caseType): Desktop
+ addLaptop(actorUserID, brand, model, screenSize): Laptop
+ updateDevice(...): boolean
+ updateDesktopCaseType(...): boolean
+ updateLaptopScreenSize(...): boolean
+ deleteDevice(actorUserID, deviceID): boolean
+ getDevices(): List<Device>
+ getDeviceByID(id): Device

Technicians

+ addTechnician(...): Technician
+ updateTechnician(...): boolean
+ deleteTechnician(actorUserID, technicianID): boolean
+ getTechnicians(): List<Technician>
+ getAvailableTechnicians(): List<Technician>
+ getTechnicianByID(id): Technician

Repair jobs

+ createRepairJob(actorUserID, customerID, deviceID, reportedProblem): RepairJob
+ assignNextWaitingJob(actorUserID): RepairJob
+ submitDiagnosisAndEstimate(actorUserID, repairJobID, diagnosis, repairDescription, estimatedCost): boolean
+ decideEstimate(actorUserID, repairJobID, approved): boolean
+ completeRepair(actorUserID, repairJobID, finalCost): boolean
+ claimRepairJob(actorUserID, repairJobID): boolean
+ deleteRepairJob(actorUserID, repairJobID): boolean
+ getRepairJobs(): List<RepairJob>
+ getWaitingQueue(): List<RepairJob>
+ getRepairJobByID(id): RepairJob

Payments

+ createPayment(actorUserID, repairJobID, amountPaid, paymentDate): Payment
+ markPaymentPaid(actorUserID, paymentID): boolean
+ getPayments(): List<Payment>
+ getPaymentByID(id): Payment
+ getPaymentForRepairJob(repairJobID): Payment

Private helpers

- find..., copy..., require..., generate..., hashPassword, verifyPassword (validation, ID generation, copying and password hashing)

dataManager 1 — *each screen holds a reference*

DashboardPanel

extends JPanel

- dataManager: DataManager
- authenticatedUser: SystemUser

+ DashboardPanel(dataManager: DataManager, authenticatedUser: SystemUser)

CustomerPanel

extends JPanel

- dataManager: DataManager
- authenticatedUser: SystemUser

+ CustomerPanel(dataManager: DataManager, authenticatedUser: SystemUser)

DevicesPanel

extends JPanel

- dataManager: DataManager
- authenticatedUser: SystemUser

+ DevicesPanel(dataManager: DataManager, authenticatedUser: SystemUser)

RepairJobsPanel

extends JPanel

- dataManager: DataManager
- authenticatedUser: SystemUser

+ RepairJobsPanel(dataManager: DataManager, authenticatedUser: SystemUser)

TechnicianPanel

extends JPanel

- dataManager: DataManager
- authenticatedUser: SystemUser

+ TechnicianPanel(dataManager: DataManager, authenticatedUser: SystemUser)

PaymentPanel

extends JPanel

- dataManager: DataManager

+ PaymentPanel(dataManager: DataManager)

UsersPanel

extends JPanel

- dataManager: DataManager
- authenticatedUser: SystemUser

+ UsersPanel(dataManager: DataManager, authenticatedUser: SystemUser)

«create»

DataManager 1 — 0..*

Domain classes (see Domain Class Diagram)

Customer, Device (Laptop, Desktop), Technician, RepairJob, Payment, SystemUser